

By: Jameel Abbas

My name is **Jameel Abbas**, and I am currently in the final stages of my **BS in Computer Science** at the University of Baltistan, Skardu. I am applying for the **Master of Science and Technology Program in Cybersecurity at École Polytechnique** because it aligns perfectly with my academic background, professional interests, and long-term aspirations of contributing to secure digital infrastructures in Pakistan and beyond.

Academic Background & Motivation

Growing up in a remote region with limited access to higher education, I developed a strong passion for computer science early in my academic journey. Despite socio-economic challenges, I have consistently maintained strong academic performance while building practical skills through part-time work as a **.NET and React.js full-stack developer**.

Through these experiences, I became increasingly aware of the growing need for robust cybersecurity practices—particularly in countries where digital transformation is expanding rapidly but security awareness and expertise are still developing. This realization motivated me to pursue advanced studies in cybersecurity.

Why École Polytechnique — MSc&T Cybersecurity

École Polytechnique's MSc&T Cybersecurity program stands out for its rigorous academic curriculum integrating **cryptography, network security, systems security, threat analysis, and applied research**. The opportunity to learn from world-class faculty, collaborate with international peers, and engage in hands-on research strongly resonates with my desire to deepen my technical and theoretical knowledge.

The interdisciplinary approach of the program—combining computer science, mathematics, and engineering—offers the ideal environment for me to grow as a researcher and security professional. I am particularly interested in exploring areas such as secure software design, vulnerability analysis, AI-driven security systems, and cyber defense strategies.

Research Interest & Professional Experience

During my undergraduate studies, I have worked on academic and professional projects involving full-stack development, REST APIs, image processing, mobile applications, and .NET Core systems. These experiences exposed me to real-world security challenges such as:

- Securing API endpoints
- Preventing unauthorized access to multi-role systems
- Understanding vulnerabilities in authentication flows
- Exposure to basic cryptographic practices
- Handling secure data in enterprise applications

These challenges sparked my interest in understanding security from a deeper, scientific perspective. I now wish to transition from being a developer who implements security features to being a professional who designs, analyzes, and ensures the security of complex systems.

Future Goals

My long-term goal is to become a cybersecurity specialist and researcher, working on:

- Designing secure systems and architectures
- Developing advanced intrusion detection and prevention mechanisms
- Contributing to Pakistan's cybersecurity ecosystem
- Creating awareness and training programs for secure software development
- Pursuing further research or working in cybersecurity labs and organizations

Ultimately, I aim to take part in strengthening the cybersecurity infrastructure of developing countries, where digital growth is rapid but security expertise is scarce.

Why I Am a Strong Candidate

I believe my academic foundation, strong motivation, professional discipline, and background in software engineering make me a committed and capable candidate for this program. My

determination to overcome challenges and pursue excellence reflects my readiness for the academic rigor at École Polytechnique.

This master's program is an essential step toward my goal of becoming a security professional capable of contributing to global cybersecurity innovations.

Conclusion

I am enthusiastic about the possibility of joining the MSc&T Cybersecurity program at École Polytechnique. With its world-class academic environment and research-driven approach, I am confident it will help me unlock my full potential and equip me with the expertise needed to make meaningful contributions to the cybersecurity community.

Thank you for considering my application. I look forward to the opportunity to be part of École Polytechnique's vibrant academic and research community.

Sincerely,

Jameel Abbas